Date: 14/4/25

Divergence

Day: M T W T F S

If $F = Pi + Qj + Rk$ is a vector field on $\mathbb{R}^3$ and $\frac{\partial P}{\partial x}$, $\frac{\partial Q}{\partial y}$ and $\frac{\partial Q}{\partial z}$ exist, then the divergence of F is the function of three variables defined by:

$$\boxed{\text{div } F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}}$$

Exp:- If $F(x,y,z) = xzi + xyzj - y^2k$, Find div F.

Solution:-

$$\text{Div } F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

$$\boxed{\text{Div } F = z + xz - 0}$$

$P = xz = z$

$Q = xyz = xz$

$R = -y^2 = -0$